

1. Primary Physical Driving Input (Fluxes vs. State Gradients)

- `plot_obukhov_heatmaps.jl` (**Flux-Driven**):
- **Input Mechanism:** Ingests surface turbulent flux measurements—specifically friction velocity (u_*) and surface heat flux (H_s or $\overline{w'\theta'}$).
- **Stability Calculation:** Directly evaluates the Monin–Obukhov length scale $L = -\frac{u_*^3 \theta_0}{\kappa g w' \theta'}$ and computes the similarity coordinate $\zeta(z, t) = z/L(z, t)$ directly from turbulent transport quantities.
- `plot_stability_heatmaps.jl` (**Gradient-Driven**):
- **Input Mechanism:** Ingests profile state gradients—specifically vertical wind shear (S) and potential temperature stratification ($\partial\theta_v/\partial z$) extracted from tower soundings or Single-Column Model (SCM) fields.
- **Stability Calculation:** Evaluates the local gradient Richardson number profile $Ri_g(z, t) = \frac{g}{\theta_0} \frac{\partial\theta_v/\partial z}{S^2}$ and **inverts** $Ri_g \rightarrow \zeta$ through a similarity closure model.

2. Mathematical Closures & Inversion Architecture

- `plot_obukhov_heatmaps.jl` (**Direct / Classical Businger–Dyer**):
- Evaluates $\zeta = z/L$ directly from surface fluxes without an inversion step.
- When profile inversions are evaluated, rational Businger–Dyer mappings ($\zeta = \frac{Ri_g}{1-5Ri_g}$) require hard numerical caps (e.g., $Ri_g \leq 0.19$) near critical thresholds ($Ri_c \approx 0.20$), flatlining ζ and artificially truncating hyper-stable nocturnal structure aloft.
- Panel 2 carries the descriptor `$\zeta(z,t)$ [Direct]` (or un-tagged) to signify pure flux-based scaling rather than a model inversion.
- `plot_stability_heatmaps.jl` (**Non-Singular GLGS / eMOST Inversion**):
- Employs SHEBA-calibrated **Grachev et al. (2007) / GLGS** flux-profile relationships:

$$\phi_m(\zeta) = 1 + \frac{a_m \zeta}{(1 + b_m \zeta)^{2/3}}, \quad \phi_h(\zeta) = Pr_0 \left(1 + \frac{a_h \zeta}{1 + b_h \zeta} \right)$$

- Uses a 1D Newton-Raphson solver (`zeta_from_rig_glgs`) to invert $Ri_g \rightarrow \zeta$ continuously across hyper-stable regimes ($0 \leq \zeta < 100$) without numerical clipping or pole singularities. Panel 2 is explicitly titled `$\zeta(z,t)$ [GLGS]` to denote closure model inversion.
- **Visual Saturation Caveat:** While the Newton-Raphson solver converges up to $\zeta < 100$ without

numerical clipping, the colormap scaling $\text{sgn}(\zeta) \log_{10}(1 + |\zeta|)$ capped at ± 1.5 saturates visually at $\zeta \approx 30.62$ ($\log_{10}(1 + 30.62) = 1.5$). Calculated values in $30.6 \leq \zeta \leq 100$ render as saturated color extrema, creating visual truncation aloft despite continuous numerical output.

3. Vertical Domain Extent and Grid Resolution

- `plot_obukhov_heatmaps.jl` : Standardized on a vertical evaluation grid of $z \in [1.0, 100.0 \text{ m}]$ at 1-meter increments (`COMMON_Z_GRID = 1.0:1.0:100.0`). Designed primarily for surface-layer and tower-footprint scaling.
- `plot_stability_heatmaps.jl` : Expanded to $z \in [1.0, 200.0 \text{ m}]$ at 1-meter increments (`COMMON_Z_GRID = 1.0:1.0:200.0`). Captures elevated Low-Level Jet (LLJ) cores, jet-driven shear erosion, and nocturnal boundary layer capping inversions (e.g., across the 200m Cabauw mast in GABLS3).

4. Derivative Stencils & Track A Regularization

- `plot_obukhov_heatmaps.jl` : Interpolates raw profiles onto a uniform 1m grid *before* applying 3-point uniform finite-difference operators (`uniform_gradient_1d` , `uniform_hessian_1d`).
- *Artifact Note*: Piecewise-linear interpolation prior to second differentiation forces $\zeta_{zz} = 0$ between sensor levels, producing horizontal "banded" Hessian zeroing artifacts.
- `plot_stability_heatmaps.jl` (**Track A Compliant**): Implements `assemble_derivatives_track_a` .
- Constructs Taylor-Vandermonde derivative matrices ($\mathbf{D}_1, \mathbf{D}_2$) directly on the **native non-uniform tower levels** in log-height space ($\xi = \ln(z/z_0)$) before projecting smooth derivative fields onto the evaluation grid.
- Enforces the **Mandatory** $N_z \geq 3$ **Stencil Gate**; sparse datasets failing this gate (e.g., 2-level SHEBA) trigger a `StencilCollapseException` and are safely routed to a Bulk Richardson (Ri_b) fallback engine (`assemble_bulk_fallback`).

5. Summary of the 4 Diagnostic Panels

Both scripts output a 2×2 panel matrix per campaign:

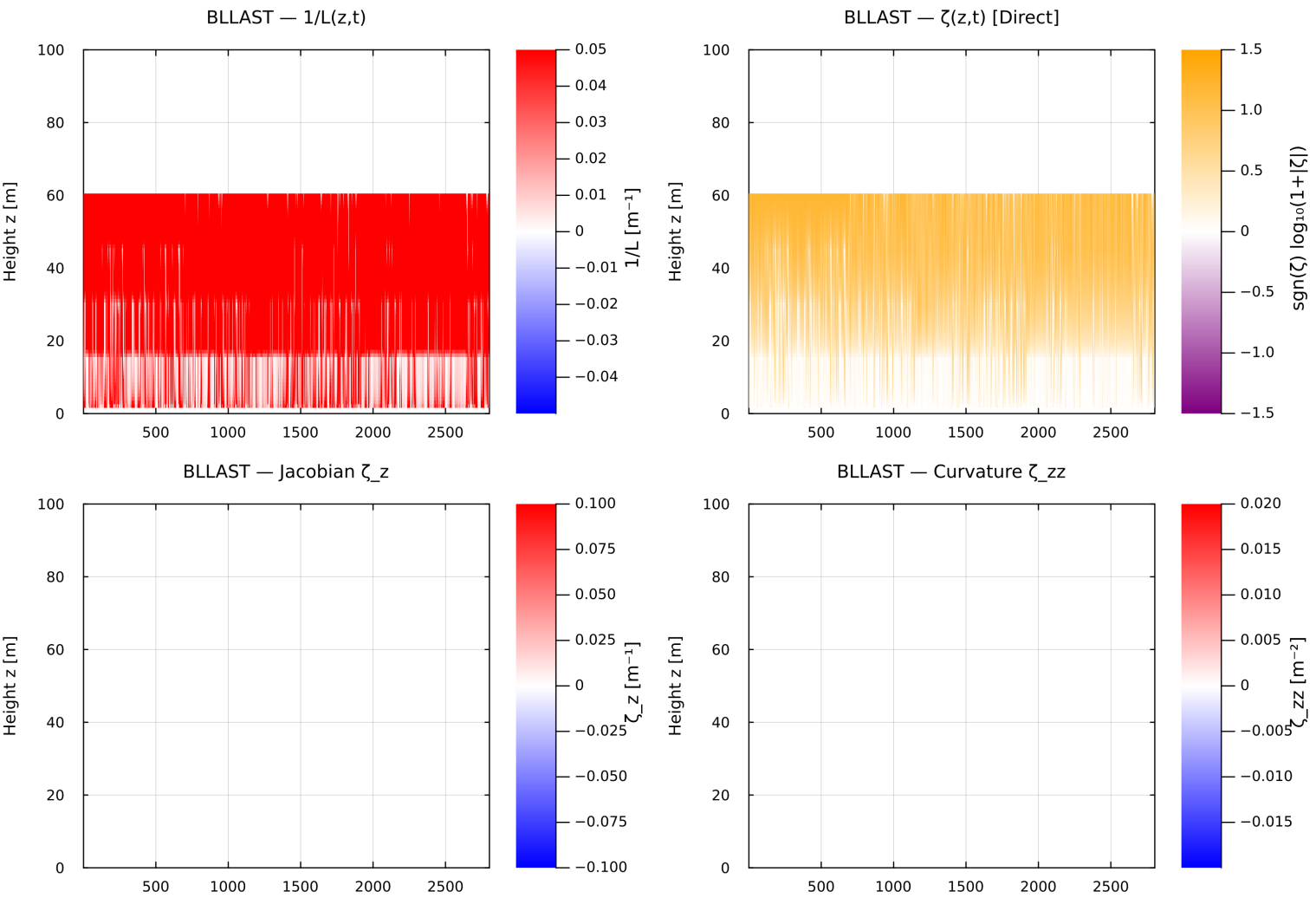
Panel	Metric	Mathematical Formulation	Visual Target & Diagnostic Role
Panel 1	Reciprocal Obukhov Length	$1/L(z,t) \quad [\text{m}^{-1}]$	Symmetric :coolwarm / :bwr colormap ($\pm 0.05 \text{ m}^{-1}$). Distinguishes convective instability ($1/L < 0$) from stable stratification ($1/L > 0$).
Panel 2	Stability Parameter	$\zeta(z,t) = z/L(z,t)$	Symmetric-log transform ($\text{sgn}(\zeta) \log_{10}(1 + \ \zeta\)$) on a :puor colormap (± 1.5). Titled [Direct] for flux data and [GLGS] for inversions. Visually saturates above $\zeta \approx 30.6$.
Panel 3	Coordinate Jacobian	$\zeta_z = \partial\zeta/\partial z \quad [\text{m}^{-1}]$	Centered stencil first derivative. Superimposes a solid white contour line at $\zeta_z = 0$ to explicitly map spatial coordinate fold loci (e.g., LLJ nose axes).
Panel 4	Coordinate Curvature	$\zeta_{zz} = \partial^2\zeta/\partial z^2 \quad [\text{m}^{-2}]$	Centered stencil second derivative ($\pm 0.02 \text{ m}^{-2}$). Maps inversion capping heights, shear erosion boundaries, and near-surface inflection masking zones.

Multi-Campaign Behavior

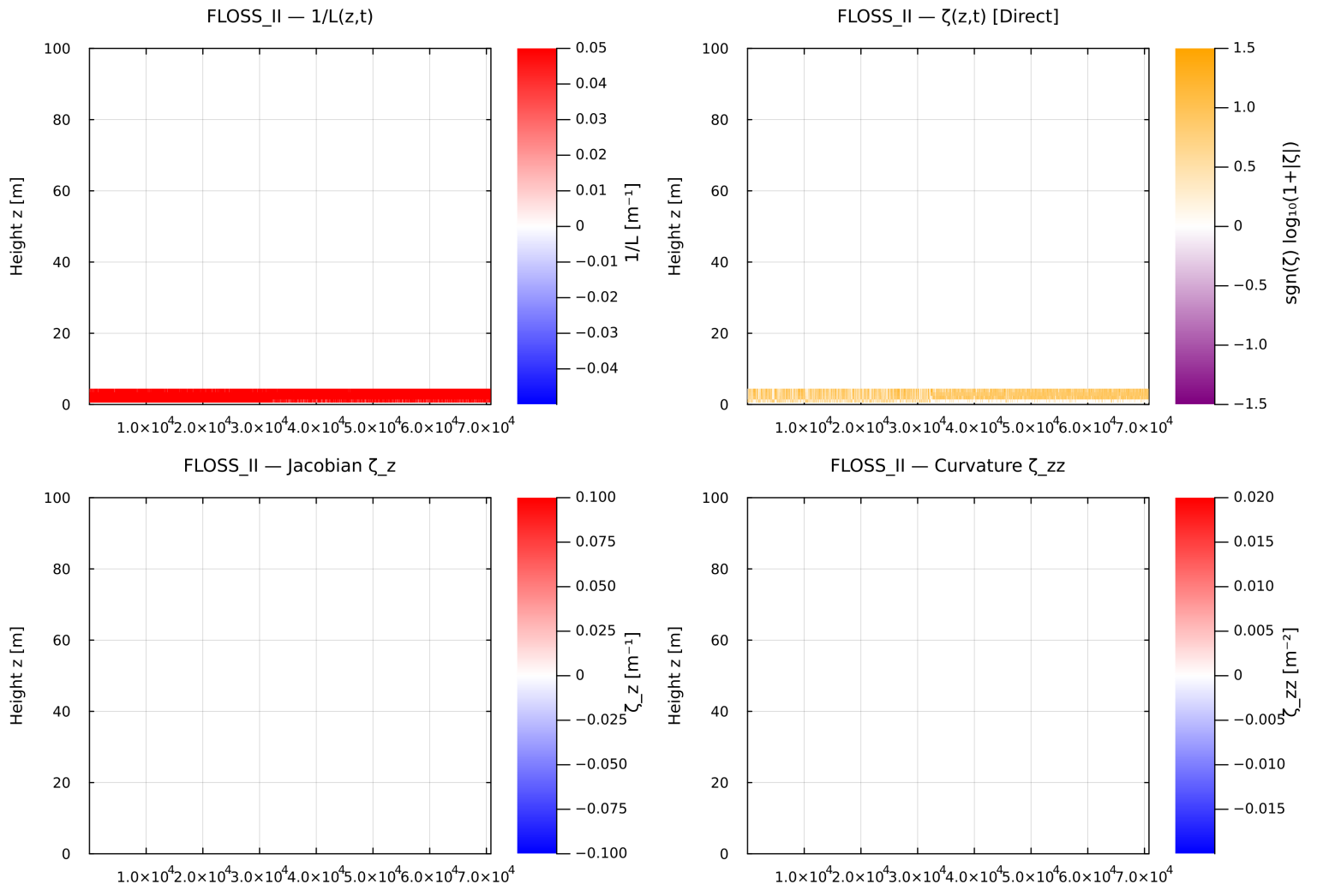
Evaluating these two scripts across campaign archetypes illustrates key resolution thresholds:

- **CASES-99** ($N_z = 7, z \leq 55 \text{ m}$): Cleanly displays the descending nocturnal Low-Level Jet nose. Panel 3's white contour ($\zeta_z = 0$) tracks the coordinate fold locus tracking the jet core.
- **GABLS3** ($N_z = 38, z \leq 200 \text{ m}$): Captures the full 24-hour diurnal cycle over the Cabauw mast, showing daytime convection ($\zeta < 0$) transitioning into elevated nocturnal shear erosion.
- **SHEBA** ($N_z = 2, z \in \{2.5, 10\} \text{ m}$): Triggers the $N_z \geq 3$ stencil gate (Bypassed Guard), routing to the 1D bulk fallback layer to avoid finite-difference under-determination.

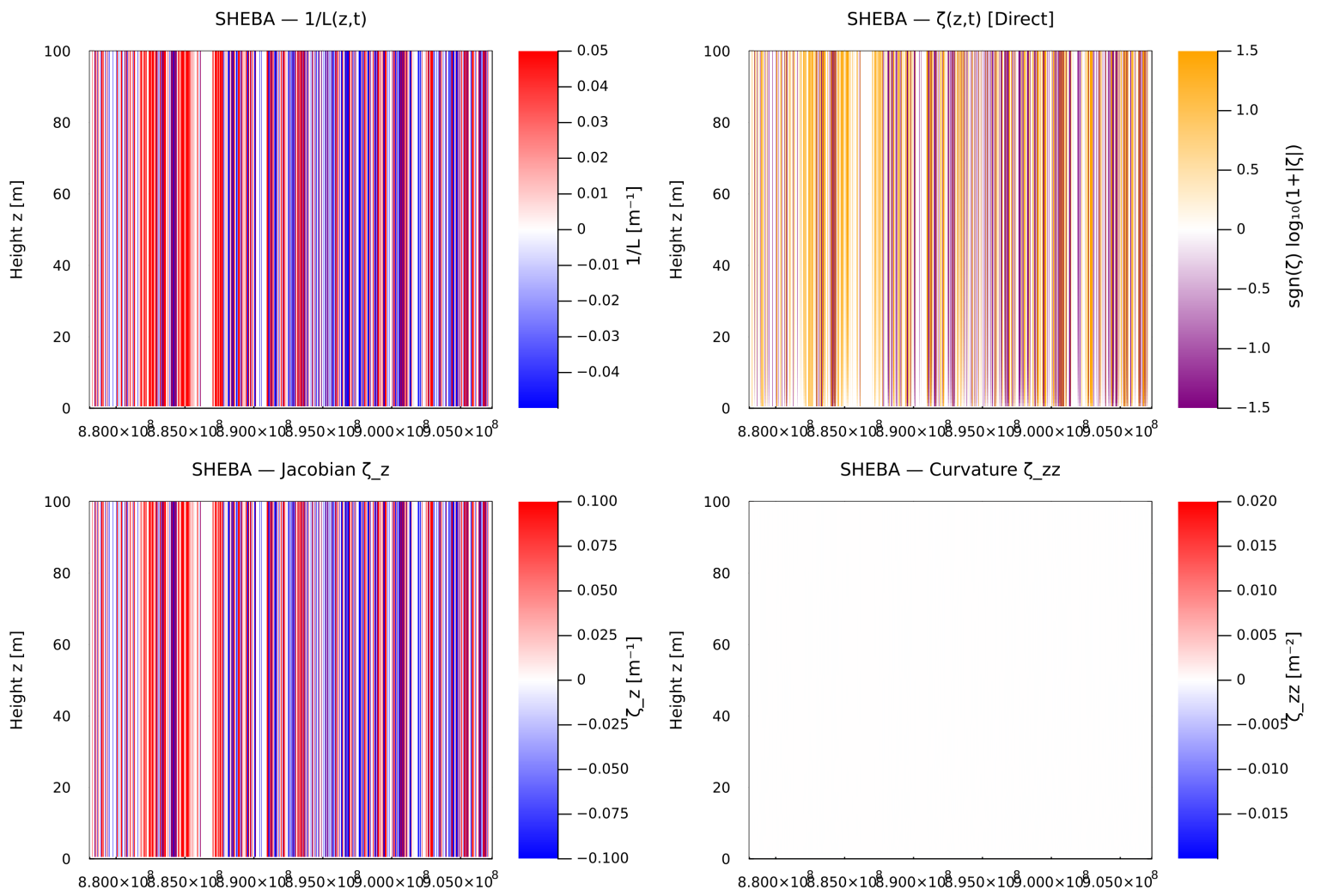
BLLAST (Flux-Driven Obukhov Diagnostic)



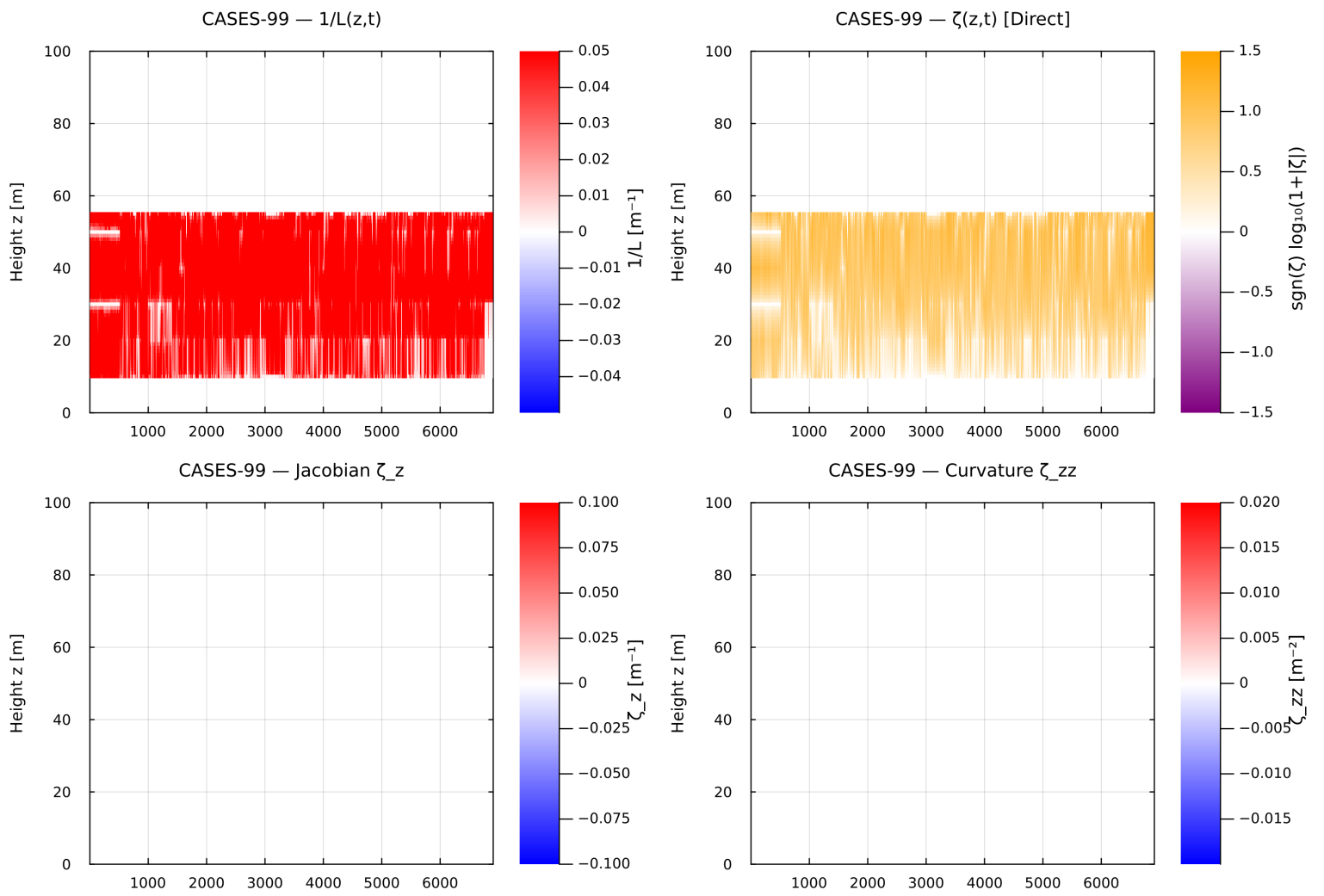
FLOSS II (Flux-Driven Obukhov Diagnostic)



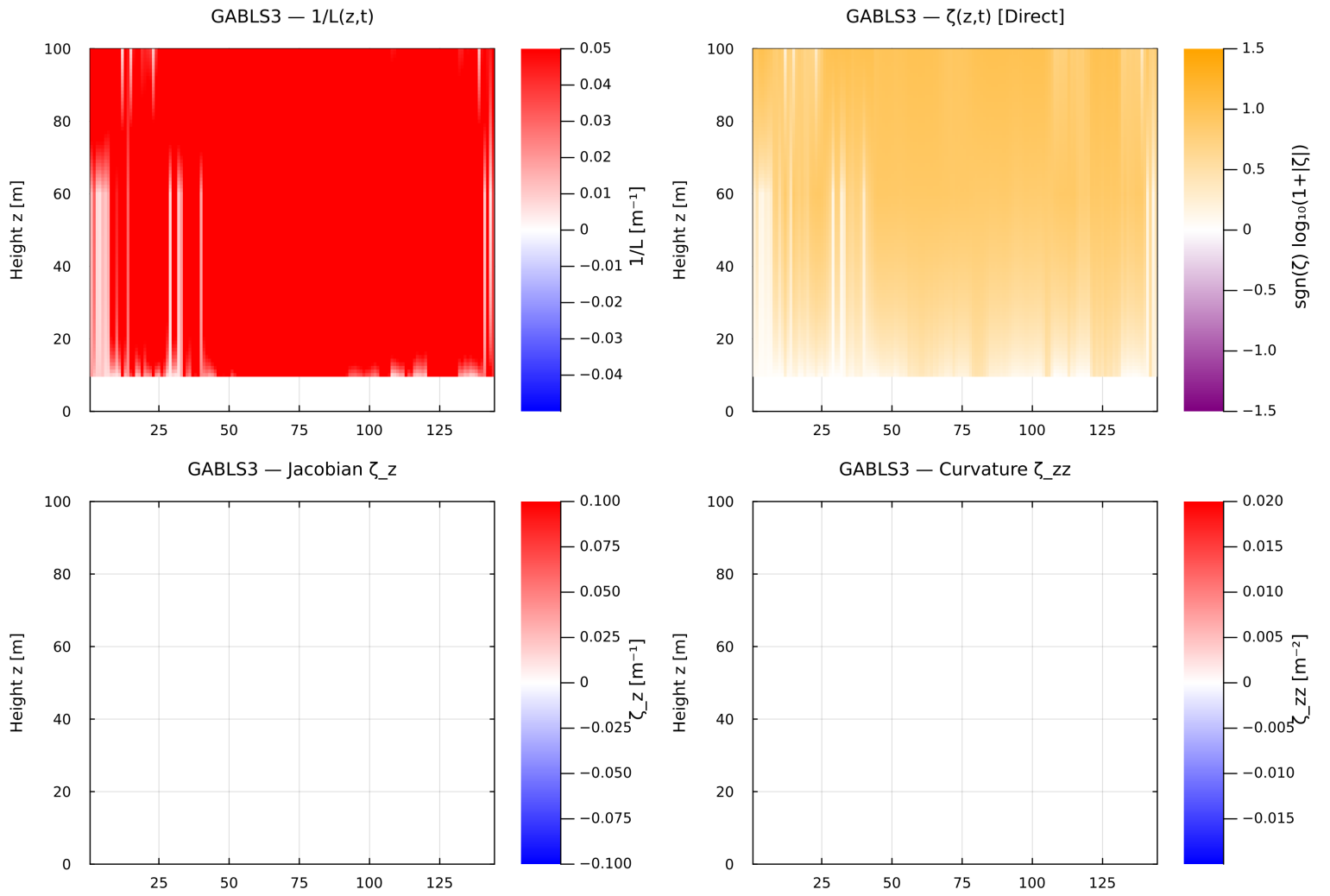
SHEBA (Flux-Driven Obukhov Diagnostic)



CASES-99 (Flux-Driven Obukhov Diagnostic)



GABLS3 (Flux-Driven Obukhov Diagnostic)



CODE

Refined Architectural Summary

1. Standardized Obukhov Heatmap Engine (`plot_obukhov_heatmaps.jl`)

Interpolates heterogeneous tower observations onto a standard grid ($z \in [1, 100]$ m) at 1 m resolution (`COMMON_Z_GRID`).

- **$1/L(z, t)$ Reciprocal Length:** Bounded to $[-0.05, 0.05] m^{-1}$ using a symmetric red-white-blue colormap (`:bwr`) to separate convective instability ($1/L < 0$) from stable boundary layers ($1/L > 0$).
- **$\zeta(z, t)$ Symmetric-Log Stability:** Employs $\text{sgn}(\zeta) \log_{10}(1 + |\zeta|)$ on a Purple-White-Orange palette (`:puor`) across $[-1.5, 1.5]$ to resolve surface layer gradients alongside hyper-stable aloft conditions.
- **Coordinate Jacobian ($\zeta_z = \partial\zeta/\partial z$):** Superimposes a solid white contour overlay at $\zeta_z = 0$ to

identify coordinate fold locations.

- **Coordinate Curvature** ($\zeta_{zz} = \partial^2 \zeta / \partial z^2$): Maps vertical inversion capping layers and shear boundaries.

2. GSPT Curvature Decomposition Engine (`gspt_curvature_contour_v3.jl`)

Partitions total observed Richardson curvature Ri_{zz} into constitutive thermodynamics and spatial coordinate geometry:

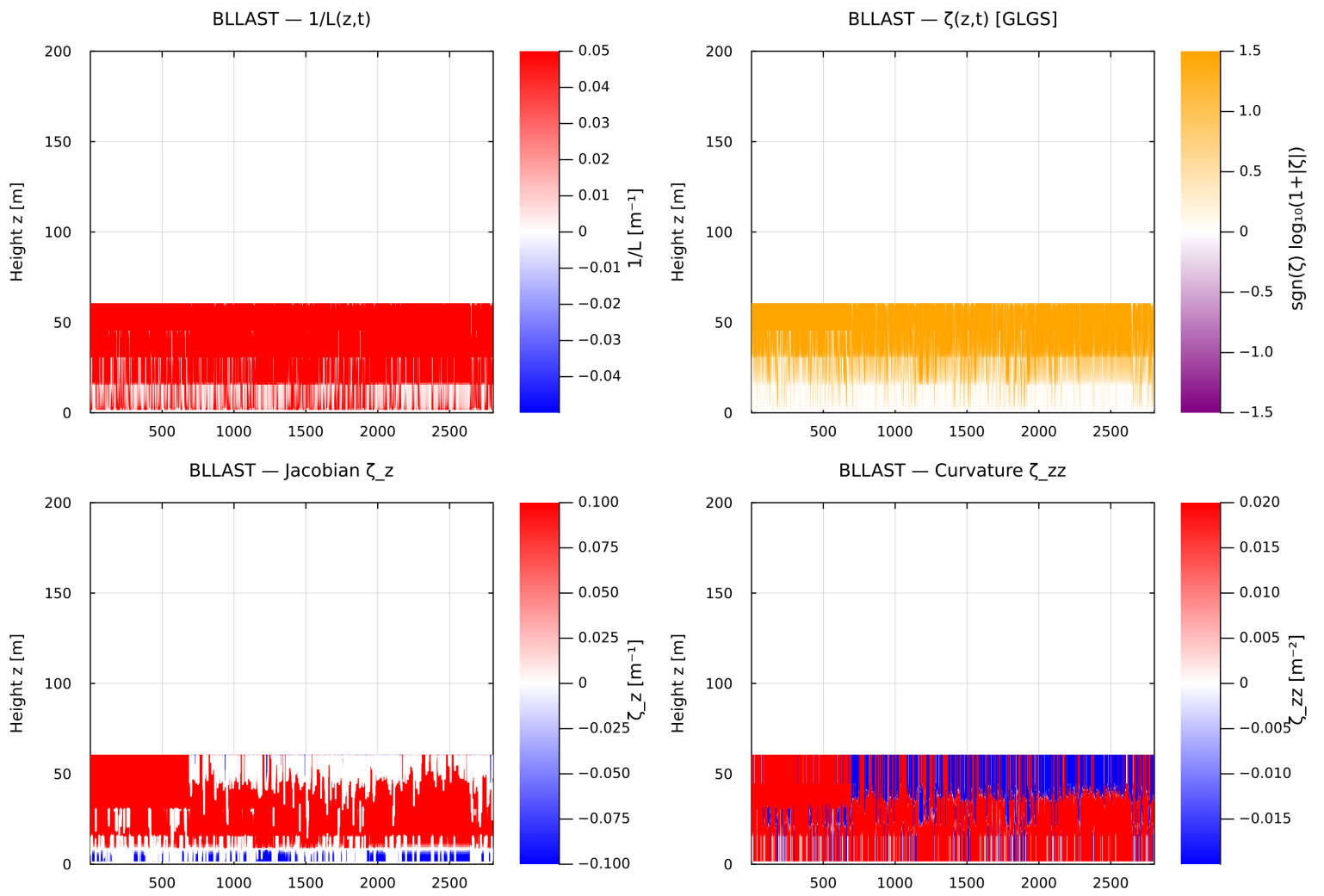
$$Ri_{zz} = \underbrace{R''(\zeta)\zeta_z^2}_{C_{\text{const}}} + \underbrace{R'(\zeta)\zeta_{zz}}_{C_{\text{coord}}} + \mathcal{E}_{\Delta z}$$

- **Non-Uniform Vandermonde Stencils** ($\mathbf{D}_1, \mathbf{D}_2$): Solves local Taylor-Vandermonde linear systems ($\mathbf{A}\mathbf{w} = \mathbf{b}$) to construct exact derivative operators for irregularly spaced tower levels ($z \in [1.5, 5.0, 10.0, 20.0, 30.0, 45.0, 55.0]$ m).
- **Tikhonov Pre-Conditioning**: Filters high-frequency observational sensor noise ($\sigma_{\text{obs}} \approx 0.008$) prior to second differentiation via $(\mathbf{I} + \lambda_{\text{reg}} \mathbf{D}_2^T \mathbf{D}_2) Ri_{\text{smooth}} = Ri_{\text{obs}}$, ensuring the residual $\mathcal{E}_{\Delta z}$ reflects spatial discretization error rather than measurement noise.

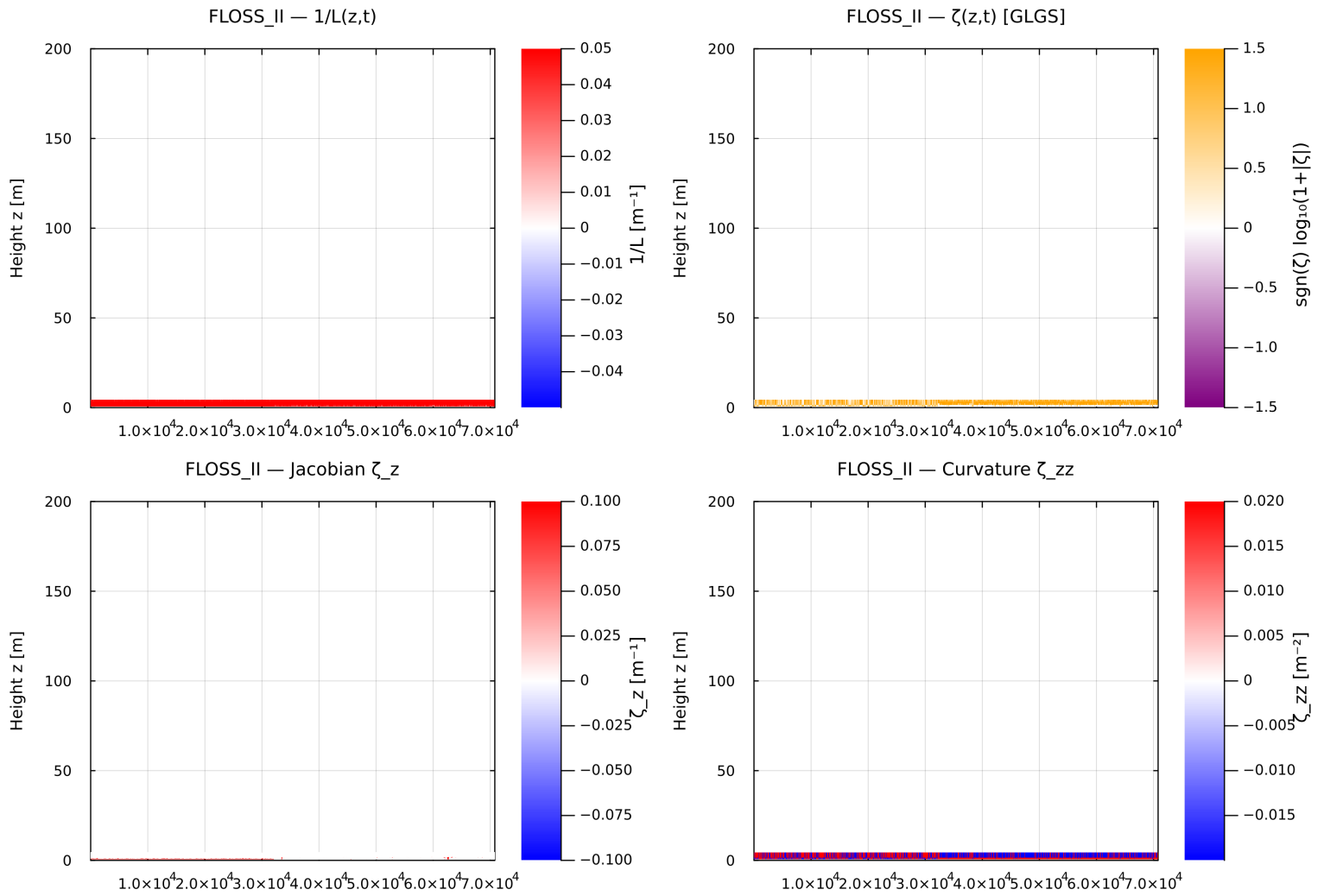
3. Primary Physical Diagnostic Takeaways

- **Inflection Masking** ($z \approx 10$ m): Positive coordinate curvature ($C_{\text{coord}} \approx +0.0036 \text{ m}^{-2}$) cancels negative stability curvature ($C_{\text{const}} \approx -0.0040 \text{ m}^{-2}$), hiding active, opposing physical mechanisms behind an apparently linear profile ($Ri_{zz} \approx 0$).
- **Jet-Nose Fold Illusion** ($z \approx 45$ m): Near Low-Level Jet (LLJ) wind maxima where vertical shear vanishes ($S^2 \rightarrow 0$), visual profile knees are driven over 99.4% by coordinate-stretching geometry (C_{coord}) rather than true turbulence collapse.

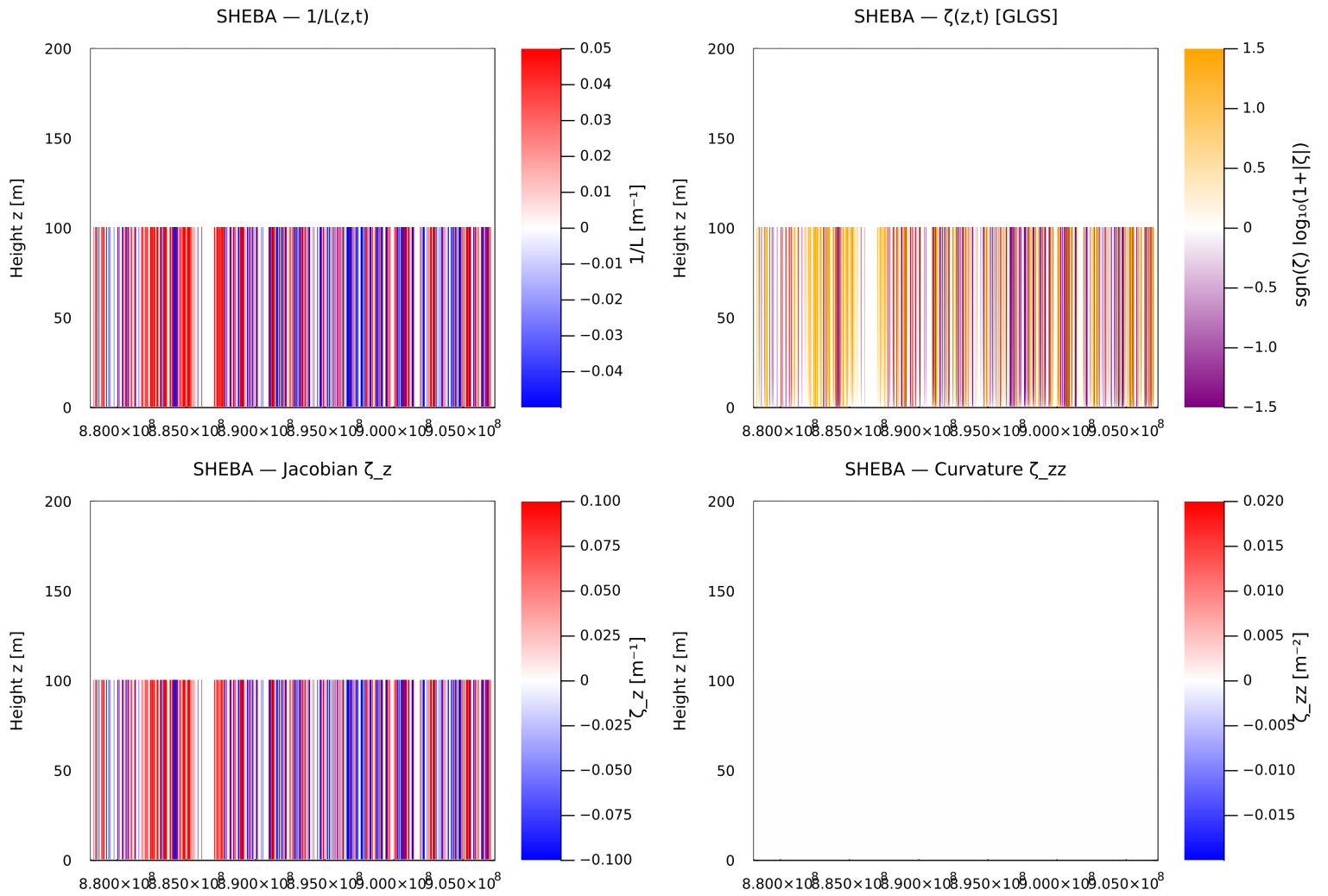
BLLAST (Gradient-Driven Stability Inversion)



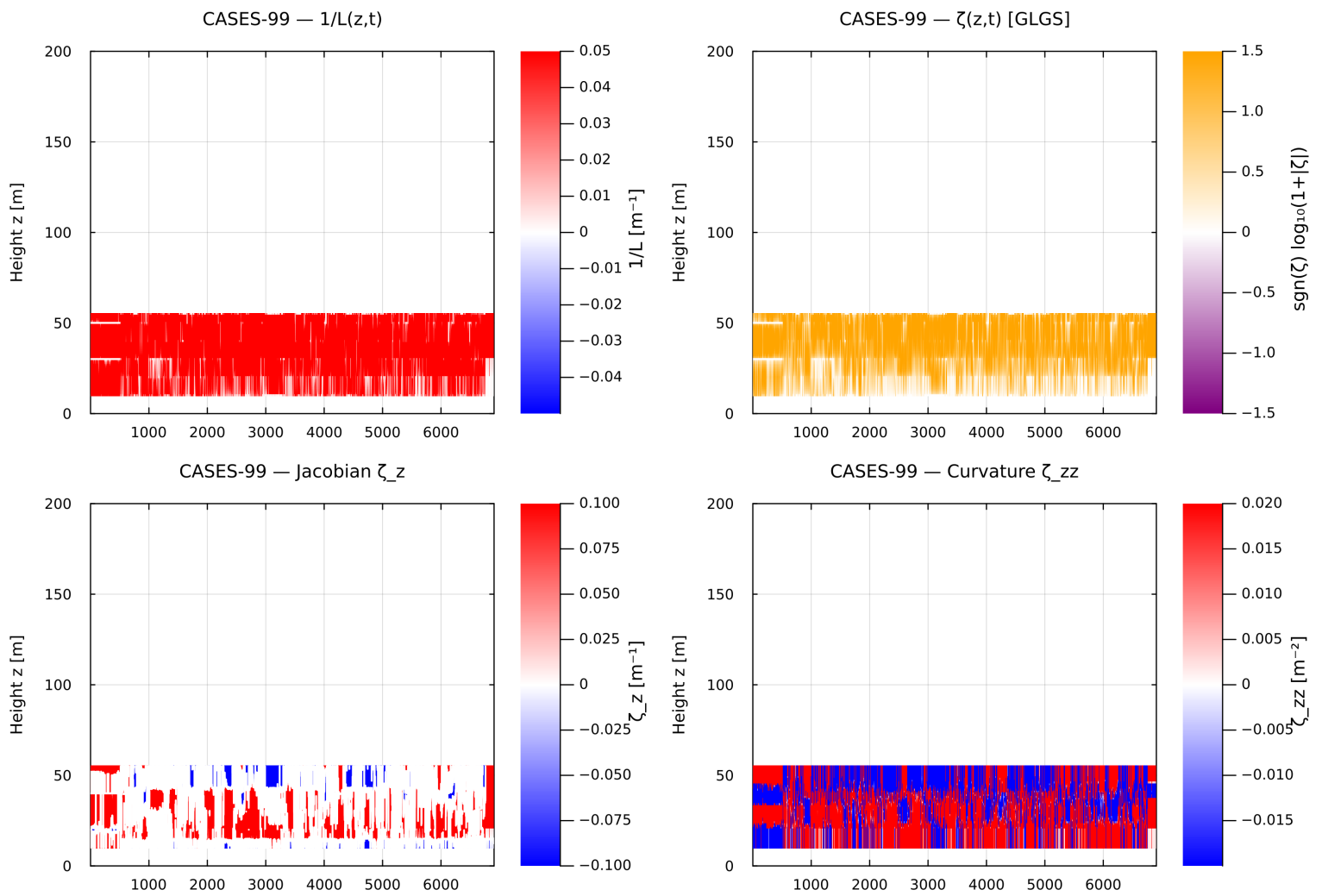
FLOSS II (Gradient-Driven Stability Inversion)



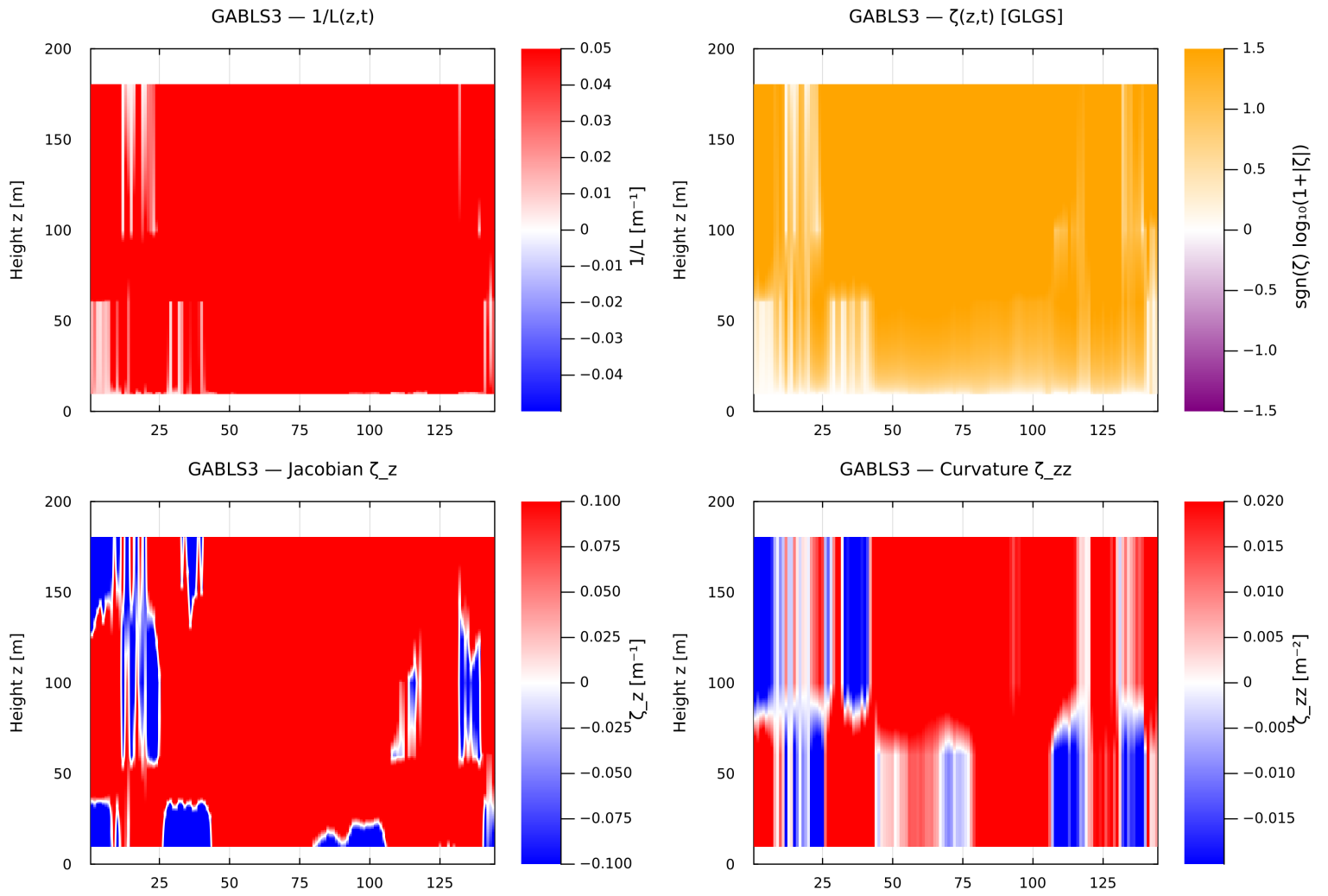
SHEBA (Gradient-Driven Stability Inversion)



CASES-99 (Gradient-Driven Stability Inversion)



GABLS3 (Gradient-Driven Stability Inversion)



[Code Reference](#)